Haiwei Yu

Applied AI / LLM Systems / AI Agents / AI Solutions

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I'm a computer science master's student at Duke. I build AI systems for practical workflows, with recent work in natural language access to health data, agent evaluation, and mining safety planning.

Education

Duke University - M.S. in Computer Science

Expected May 2027 | GPA: 3.90/4.00 | University

The University of Utah - B.S. in Computer Science

Jan 2021-May 2025 | GPA: 3.77/4.00 | University

Dean's List: Spring 2021, Fall 2021, Fall 2022, Fall 2023, Spring 2024, Fall 2024

Research Experience

Research Assistant - AI Systems for Health Data

Duke University | Jan 2026-Present

- Design and develop an LLM-based system that helps users query electronic health records in natural language, with validation and traceable query execution.
- Independently developed the system, conducted experiments and failure analysis, and prepared the first-author manuscript.
- Contributed to system design for a separate collaborative study on agent-based natural language database querying.

Publications and Manuscripts

Reliable and Auditable EHR Querying via Semantic IR and Deterministic SQL Compilation

Haiwei Yu; Jian Lu; Raymond M Xiong; Danyang Zhuo; Anru Zhang

Work in progress

Developing an LLM-based system for natural language access to electronic health records, with a focus on reliable query execution and auditability. I independently carried out the system implementation, experiments, and manuscript preparation.

Iteration Without Elaboration: A Simple ReAct Architecture Suffices for Text-to-SQL Generation

Jian Lu; Haiwei Yu; Raymond M Xiong; Anru Zhang; Danyang Zhuo

Public preprint; under review | Preprint first posted 2026-08-23 | arXiv

Collaborative research on iterative agent-based database querying. Contributed to system design for an LLM-based agent that queries databases through iterative interaction.

Leveraging Prompt Engineering to Facilitate AI-driven Chart Generation

Haiwei Yu

Published | 2024 · CISAI 2024 · pp. 261-268 | DOI

Sole-author research on prompt engineering for AI-driven chart generation. Independently carried out the research and prepared the manuscript.

Teaching and Service

Teaching Assistant - CS 4400 Computer Systems

The University of Utah | Aug-Dec 2024

- Led lab sessions, answered course and assignment questions, and graded student work with feedback.

Course syllabus (PDF)

Teaching Assistant - CS 5530 Database Systems

The University of Utah | Jan-May 2025

- Supported multi-stage database projects, held help hours, and graded assignments, exams, and project submissions.

Course syllabus (PDF)

Invited Reviewer - IEEE ICHI 2026

IEEE International Conference on Healthcare Informatics | 2026

- Invited to review for IEEE ICHI 2026.

Conference website

Selected Projects

SafeStart

Duke University | Sep 2026-Present

- Developing an AI assistant for mining pre-task planning that uses safety requirements, SOPs, and incident references to help draft job safety analyses, with supervisor review.
- Developing retrieval-assisted drafting, missing-context checks, and an explicit supervisor review workflow.

PolyglotDRA-SecBench

Duke University | Mar-May 2026

- Built an offline English/Chinese benchmark for deep-research agents under clean and adversarial retrieval conditions.
- Integrated MiroThinker and Tongyi DeepResearch, normalized tool-use traces, and automated scoring of task utility and undesirable agent behavior, including goal drift, decoy engagement, and taint leakage.

QueueHub (QHub)

Duke University | Aug 2025-May 2026

- Built a real-time TA queue with role-based access and atomic claim locking to coordinate concurrent teaching-assistant actions.
- Translated instructor and TA workflows into wait-time and workload analytics, including P50/P95 statistics, and containerized the application for a departmental pilot.

UParkNow

The University of Utah · Capstone | Aug 2024-May 2025

Independent backend design and implementation; the full product was a team capstone

- Designed and implemented the Java Spring Boot backend, including REST APIs, authentication, data persistence, and mail integration.
- Built WebSocket communication to synchronize parking updates between the recognition system, Android clients, and the management interface.
- Deployed the backend on AWS EC2 with MariaDB on Amazon RDS and wrote 16 automated backend tests.

Backend repository | Design document | Demo video

Internship Experience

Applied Machine Learning Intern

Institute of Automation, Chinese Academy of Sciences · Beijing | Jul-Aug 2023

- Built Python and OpenCV workflows for image collection, annotation, preprocessing, and quality checks.
- Reproduced and fine-tuned YOLOv5/YOLOv8 models, testing data augmentation and training settings to improve detection performance and inference efficiency.

Java Development Intern

Gansu Haorun Technology Information Co., Ltd. · Gansu | Jul 2020-Jul 2021

- Developed and debugged Java/Spring backend components for an internal attendance management system, working with MySQL data storage and SQL access.
- Participated in code reviews, functional testing, and issue resolution during system delivery.

Research Training

Deep Learning for Natural Language Processing

Short research training | Aug 2024

- Studied neural language models and Transformer architectures through a short research training project.
- Tutor associated with Johns Hopkins University.

Geometry Processing for 3D Reconstruction

Short research training | May-Jun 2022

- Implemented point-cloud registration and mesh-processing experiments in C++, using ICP, RANSAC, filtering, and nearest-neighbor search with OpenMesh and ANN.
- Tutor associated with the University of California, Berkeley.

Technical Skills

AI systems: Large Language Models (LLMs), AI Agents, Retrieval-Augmented Generation (RAG), Agent Evaluation, Natural Language Processing (NLP), Text-to-SQL, Prompt Engineering, Tool Use

AI workflows & evaluation: Workflow Design, Human Review, Model Evaluation, Error Analysis

Programming: Python, Java, SQL, TypeScript / JavaScript, C++

ML & data: PyTorch, scikit-learn, Pandas, Polars, Spark / PySpark, SQL, OpenCV

Backend & application systems: Spring Boot, Spring Security, REST APIs, WebSocket, JPA / Hibernate, MySQL / MariaDB, MongoDB, Nuxt, React, Next.js

Delivery: Docker, Git / GitHub, CI/CD, AWS EC2 / RDS, Kubernetes (QueueHub)